F79SP Session 2: Introduction to Random Walks

1 More on Limit Theorems

Theorem 1.1 (Central Limit Theorem:)**.** *Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s.*

1. *Mean $\mu = \mathbf{E}X_1$ and variance $\sigma^2 = \mathbf{Var}X_1$ are finite.*
2. *Let $S_n = X_1 + X_2 + \cdots + X_n$ and Y be a standard normal random variable.*

Then, when $n \to \infty$, we have:

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} = \frac{\frac{S_n}{n} - \mu}{\sqrt{\frac{\sigma^2}{n}}} \to_D Y.$$

Remark 1.1. *CLT implies WLLN.*

- By CLT, $\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \to_D Y \sim N(0,1)$.
- $\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \times \frac{\sigma}{\sqrt{n}} \to_D Y \times 0$ (Using $lim_{n\to\infty}(F(x)G(x)) = lim_{n\to\infty}F(x)lim_{n\to\infty}G(x)$)
- $\frac{S_n - n\mu}{n} \to_D 0$
- $\frac{S_n - n\mu}{n} \to_P 0$ (Using the fact that $\to_D c$ and $\to_P c$ for a constant c are equivalent)
- $\frac{S_n}{n} \to_P \mu$.

2 Random walks

Let's go into definition of random walks.

Definition 2.1 (Random Walks)**.** *Let $a \in \mathbb{R}$ and $\xi_1, \xi_2, \ldots$ be i.i.d. r.v's. The discrete-time stochastic process $\{S_n\}_{n\geq 0}$ defined as:*

$$S_0 = a, S_n = a + \sum_{i=1}^{n} \xi_i \text{ for } n > 0.$$

is called a random walk (or sometimes, drunkard's walk).

Remark 2.1. *Below are some remarks:*

1. *We can think of $S_0 = a$ as the starting point of the walk.*
2. *The i.i.d. r.v.'s ξ_i's are called increments of the underlying random walk.*
3. *Note that increments can be discrete or continuous.*

Example 2.1. *Let $a = 5$ and ξ_1 such that $\mathbf{P}(\xi_1 = 1) = 0.6$, $\mathbf{P}(\xi_1 = 2) = 0.2$ and $\mathbf{P}(\xi_1 = -1) = 0.2$. The random walk $S_n = a + \sum_{i=1}^{n} \xi_i$ can be visualised as follows:*

- $S_0 =$.
- $S_1 =$

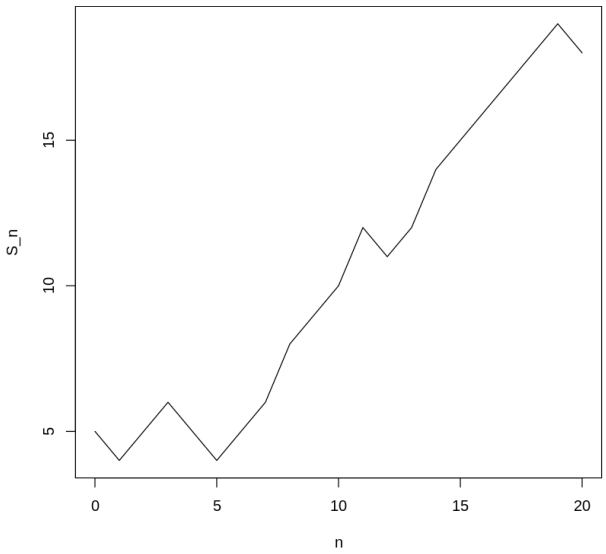


Figure 1: A realisation of the given random walk

Food for thought:

- *How would you expect S_n behaves for large n?*
- *Is it possible that $S_n \leq 0$ at $n = 5$? How about at some n?*
- *An explicit formula of S_n?*

Notes: The mean of increment $\mu = \mathbf{E}\xi_1$ governs the average motion of the walk. μ is also called the **drift** of the random walk.

Interesting ideas of random walks: Capital growth and ruin problem, the random walk hypothesis, Brownian motion etc.

Properties of random walks

We study the following properties of $\{S_n\}_{n\geq 0}$:

- $\mathbf{E}S_n$
- $\mathbf{Var}S_n$
- Its behaviour for large n.

Let

$$\mu = \mathbf{E}\,\xi_i \qquad \textbf{mean} \text{ of increments } \xi_i$$
$$\sigma^2 = \mathbf{Var}\xi_i \qquad \textbf{variance} \text{ of increments } \xi_i.$$

We shall assume that both the above quantities exist and are finite.

We have:

- $\mathbf{E}\,S_n =$

- $\mathbf{Var}S_n =$

Further, by the Central Limit Theorem (CLT), for moderate to large n, S_n is approximately normally distributed with mean and variance as given above, i.e.

$$S_n \sim N(a + n\mu,\, n\sigma^2) \qquad \text{approximately.}$$